

System vision document

Problem description:

Healthcare workers face daily challenges with confusing patient information, complicated scheduling, and communication issues between doctors, nurses, pharmacists, and administrative staff. This can lead to delays, errors in patient records, and difficulty keeping everyone on the same page. Patients, too, often struggle to access their own health information, which can reduce their involvement in their own care which leads to worry and distress. To address these issues, a system is needed to bring all health-related information and tasks into one place, making it easier for everyone to do their jobs well and provide better patient care.

System capabilities:

seamless access to Patient Records: Provide easy, real-time access for doctors, nurses, and pharmacists to view and update accurate patient records, including medical history and treatment plans, all in one place.

Patient Portal: Allow patients to see their own health records, appointments, and prescriptions and to securely message their healthcare team, giving them more control and clarity in their care.

Easy Appointment Scheduling and Reminders: Enable administrative staff to manage and schedule appointments and automatically send reminders to patients to reduce missed visits.

Claims and Billing Management: Allow insurance representatives to review and approve claims, streamlining the billing process for both the healthcare provider and patient.

Medication and Inventory Management: Allow pharmacists to fill medication orders and track stock levels, ensuring patients get the medications they need with no delay

User Stories

Healthcare management system

users / stakeholders:

- Doctor
- Patient
- Nurse
- Insurance representatives
- Pharmacists

User stories:

1. Doctor:

- “As a doctor, I want to view and update patient medical records in at any time so that I can make proper treatment decisions and consistent care”

System analyst:

- The doctor needs unrestricted, on-demand access to patient records. The doctor must be able to modify or add information to records. The system must be accessible at

any time (24/7) to support medical staff and ensure that updated information is immediately available to all relevant staff.

Acceptance criteria:

- Access to and updating of patient medical records is restricted and only authorized doctors can check.
- Before patient data can be accessed, the system validates user credentials.

2. Nurse:

- As a nurse, I want to view and record patients vitals and care notes so that I can monitor their health status and keep the healthcare team updated.

System analyst:

- must be able to record ongoing observations and retrieve patient information, which include vital signs and treatment instructions. This guarantees that other members of the healthcare team are aware of the patient's current status and allows nurses to give the most recent patient care.

Acceptance criteria:

- Real-time access to relevant patient data, including vital signs, medications, medical history, and allergies, is available to nurses.
- Nurses have the ability to enter and update patient data, including heart rate, blood pressure, and temperature.

3. Insurance representatives:

- As an insurance representative, I want to review and approve patient claims so that I can process billing accurately and ensure proper coverage.

System analyst:

- to access, examine, and authorize patient claims for billing according with the patient's coverage information. This allows for precise claim processing, guaranteeing that the patient and healthcare provider receive appropriate bills in line with insurance coverage.

Acceptance criteria:

- Each patient's claim history, including information on diagnosis, treatment, and any prior claims submissions, is visible to the representative. to give an accurate bill

4. Pharmacists:

- As a pharmacist, I want to access and fulfill patient medication orders and update inventory levels so that I can ensure patients receive the correct prescriptions and maintain accurate stock levels.

System analyst:

- to monitor the amount of inventory while reviewing, completing, and updating patient prescriptions. In addition to helping the pharmacy manage inventory to avoid shortages, this guarantees that patients receive accurate and timely prescriptions.

Acceptance criteria:

- Pharmacists can view all pending medication orders, including details such as patient name, prescribed medications, dosage, and instructions.
- The system organizes orders by urgency, showing urgent prescriptions at the top of the list

5. Patient:

- As a patient, I want to view my medical records, upcoming appointments, and prescriptions so that I can stay informed about my health and manage my care effectively.

System analysts:

- to see and monitor personal health data, such as past medical records, appointments, and current medicines. This encourages greater treatment adherence and proactive health management by enabling individuals to remain informed and involved in their own healthcare.

Acceptance criteria:

- With real-time information on any changes, patients may safely access their own medicines, upcoming appointments, and medical data. When necessary, they get alerts for prescription refills and appointment reminders. Data privacy is carefully preserved, with only authorized persons having access and encryption protecting confidentiality.
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“Brief use case description” table

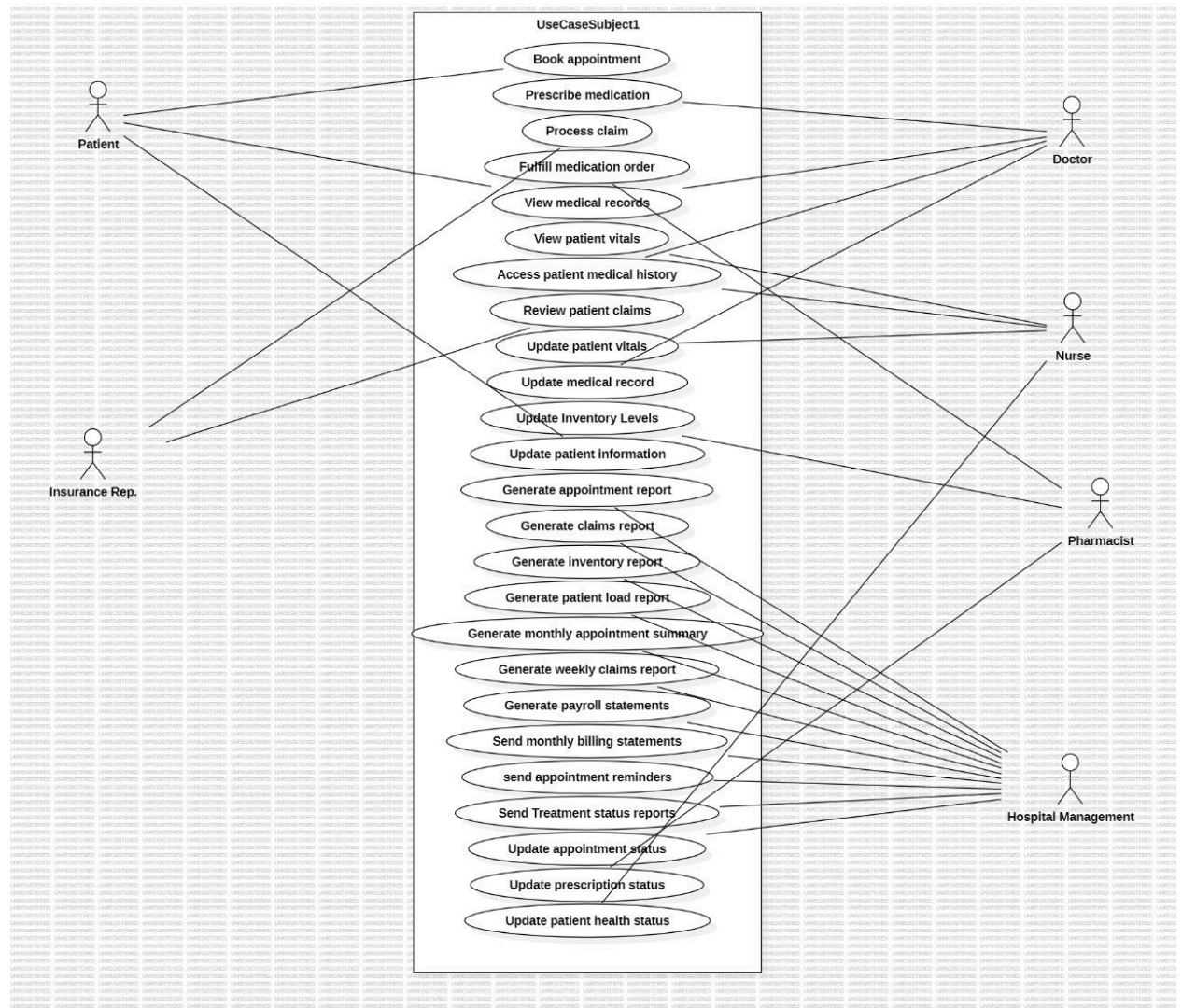
Use Case	Description
Book appointment	Patient requests to schedule a new appointment with a doctor.
Prescribe medication	Doctor prescribes medications to a patient and updates the patient’s record.
Process claim	Insurance representative reviews and processes a patient's claim for billing.
Fulfill medication order	Pharmacist reviews and dispenses medications for a patient’s prescription.
View medical records	Patient requests to view their medical history, treatments, and prescriptions.
View patient vitals	Nurse checks and reviews a patient’s vitals and care notes for monitoring.
Access patient medical history	Doctor accesses the patient's full medical history to make informed decisions.

Review patient claims	Insurance representative reviews submitted patient claims for approval.
Update patient vitals	Nurse updates a patient's vital signs and care notes in the system.
Update medical record	Doctor modifies or adds information to a patient's medical record.
Update Inventory Levels	Pharmacist updates medication inventory levels after dispensing medication.
Update patient information	Patient updates personal information such as address or phone number.
Generate appointment report	Management generates a report showing the total number of appointments.
Generate claims report	Management generates a report showing all claims processed in a specific period.

Generate inventory report	Management generates a report on current stock levels for medications.
Generate patient load report	Management generates a report to track patient load per doctor.
Generate monthly appointment summary	System automatically generates a monthly summary of patient appointments.
Generate weekly claims report	System generates a weekly report summarizing processed insurance claims.
Generate payroll statements	System generates payroll statements at the end of each payroll period.
Send monthly billing statements	System sends out billing statements to patients at the start of each month.
Send monthly billing statements	

	System sends appointment reminders to patients 24 hours before their scheduled time.
Send Treatment status reports	System sends updates to patients about their ongoing treatments.
Update appointment status	System updates the status of a patient's appointment (e.g., confirmed, canceled).
Update prescription status	System changes the status of a prescription to indicate it has been filled or refilled.
Update patient health status	System updates a patient's health record to reflect any changes in their vitals.

Use case diagram



1- External Events:

- ***External agent wants something resulting in a transaction***
 - **Event:** a patient wants to book an appointment
Use Case: book appointment
 - **Event:** a doctor wants to prescribe medication to a patient
Use Case: prescribe medication
 - **Event:** an insurance representative wants to process a claim
Use Case: process claim
 - **Event:** a pharmacist wants to fulfill a medication order
Use Case: fulfill medication order
- ***External agent wants some information***
 - **Event:** a patient wants to view their medical records
Use Case: view medical records
 - **Event:** a nurse wants to check patient vitals and care notes
Use Case: view patient vitals
 - **Event:** a doctor wants to access a patient's medical history
Use Case: access patient medical history
 - **Event:** an insurance representative wants to review patient claims
Use Case: review patient claims
- ***Data changed and needs to be updated***
 - **Event:** a nurse records updated vitals for a patient
Use Case: update patient vitals
 - **Event:** a doctor updates a patient's medical record
Use Case: update medical record
 - **Event:** a pharmacist updates the inventory after dispensing medication
Use Case: Update Inventory Levels

- **Event:** a patient updates their personal information (ex: address, phone no.)

Use Case: update patient information

- ***Management wants some information***

- **Event:** hospital management wants to see the total number of appointments scheduled

Use Case: generate appointment report

- **Event:** management wants to review the total claims processed

Use Case: generate claims report

- **Event:** management wants to assess medication stock levels

Use Case: generate inventory report

- **Event:** management wants to track patient load by doctor

Use Case: generate patient load report

2- Temporal Events:

- ***Internal output needed***

- **Event:** management requires monthly summary reports of patient appointments

Use Case: generate monthly appointment summary

Trigger: at the end of each month

- **Event:** Management needs operational reports of patient claims processed weekly

Use Case: generate weekly claims report

Trigger: at the end of each week

- **Event:** internal statements for payroll processing are needed

Use Case: generate payroll statements

Trigger: at the end of each payroll period

- ***External Outputs Needed***

- **Event:** patients need to receive monthly statements of their billing
Use Case: send monthly billing statements
Trigger: at the beginning of each month
- **Event:** appointment reminders need to be sent to patients
Use Case: send appointment reminders
Trigger: 24 hours before the scheduled appointment
- **Event:** status reports for ongoing treatments must be sent to patients
Use Case: send Treatment status reports
Trigger: upon updating treatment plans or at scheduled intervals

Step 3: Identify State Events

- *State Events*

- **Event:** a patient's appointment status changes (ex: confirmed, canceled)
Use Case: update appointment status
State Change: The system transitions from an **unconfirmed** state to **confirmed** or **canceled**
- **Event:** a prescription is filled or refilled
Use Case: update prescription status
State Change: The prescription status changes from **pending** to **filled** or **refilled**
- **Event:** a patient's health status is updated
Use Case: update patient health status
State Change: the patient's health record changes to reflect the new vital signs.

4 - Review for perfect technology assumption

The **perfect technology assumption** we will exclude any events related to system controls such as login, logout, password changes, and database backups or restores the identified external events, temporal events, and state events are all user-driven and reflect the functional requirements of the system

Event Type	user	Event Description	Use Case	Trigger/State Change
External Events	patient	A patient wants to book an appointment	book appointment	-
External Events	doctor	a doctor wants to prescribe medication to a patient.	prescribe medication	-
External Events	Insurance representative	An insurance representative wants to process a claim.	Process Claim	-
External Events	pharmacist	A pharmacist wants to fulfill a medication order.	Fulfill Medication Order	-
External Events	patient	A patient wants to view their medical records.	View Medical Records	-
External Events	nurse	A nurse wants to check patient vitals and care notes.	View Patient Vitals	-
External Events	doctor	A doctor wants to access a patient's medical history.	Access Patient Medical History	-
External Events	Insurance representative	An insurance representative wants to review patient claims.	Review Patient Claims	-
External Events	nurse	A nurse records updated vitals for a patient.	Update Patient Vitals	-
External Events	doctor	A doctor updates a patient's medical record.	Update Medical Record	-

External Events	pharmacist	A pharmacist updates the inventory after dispensing medication.	Update Inventory Levels	-
External Events	patient	A patient updates their personal information.	Update Patient Information	-
External Events	Hospital management	Hospital management wants to see the total number of appointments scheduled.	Generate Appointment Report	-
External Events	Hospital management	Management wants to review the total claims processed.	Generate Claims Report	-
External Events	Hospital management	Management wants to assess medication stock levels.	Generate Inventory Report	-
External Events	Hospital management	Management wants to track patient load by doctor.	Generate Patient Load Report	-
Temporal Events	Hospital management	Management requires monthly summary reports of patient appointments.	Generate Monthly Appointment Summary	At the end of each month
Temporal Events	Hospital management	Management needs operational reports of patient claims processed weekly.	Generate Weekly Claims Report	At the end of each week

Temporal Events	Hospital management	Internal statements for payroll processing are needed.	Generate Payroll Statements	At the end of each payroll period
Temporal Events	Hospital management	Patients need to receive monthly statements of their billing.	Send Monthly Billing Statements	At the beginning of each month
Temporal Events	Hospital management	Appointment reminders need to be sent to patients.	Send Appointment Reminders	24 hours before the scheduled appointment
Temporal Events	Hospital management	Status reports for ongoing treatments must be sent to patients.	send treatment status reports	Upon updating treatment plans or at scheduled intervals
State Events	Hospital management	A patient's appointment status changes	Update Appointment Status	The system transitions from unconfirmed to confirmed or canceled
State Events	pharmacist	A prescription is filled or refilled	Update Prescription Status	The prescription status changes from pending to filled or refilled
State Events	nurse	A patient's health status is updated	Update Patient Health Status	The patient's health record changes to reflect new vital signs